- *MCP server*—Runs independently (often over `stdio` or HTTP) and exposes callable tools. In our example, the MCP server defines the tool `get-chuck-joke`, which retrieves random facts from an external API.
- *External API or data source*—The Chuck Norris API (https://api.chucknorris.io) serves as our external data source. The MCP server queries it and returns the results to the client through the SDK and API route.

12.3.2 Building an end-to-end integration with the MCP in Next.js

We begin by copying the starter-project folder as the foundation. The main goal is to integrate MCP tools while keeping the AI model isolated from external APIs.

In this example, the MCP server runs locally and is instantiated using the Vercel AI SDK, which provides the infrastructure to expose callable tools to your AI assistant. Communication between the Next.js app and the MCP server is handled via `StdioServerTransport`, which uses standard input/output streams to send and receive structured messages. This allows the MCP server to run as a separate process on the same machine, making it easy to demonstrate tool integration without relying on network configuration. First is an MCP server implementation, as shown in the following listing.

Listing 12.1 Example MCP server for Chuck Norris jokes

```
import { McpServer } from '@modelcontextprotocol/sdk/server/mcp.js';
import { StdioServerTransport } from '@modelcontextprotocol/sdk/server/stdio.js';
import { z } from 'zod';

const server = new McpServer({
  name: 'chuck-norris-mcp',
  version: '1.0.0',
});

/ Register joke tool
server.tool(
  "get-chuck-joke",
  "Fetch a random Chuck Norris joke",
  {},
  async () => {
    try {
      const response = await fetch("https://api.chucknorris.io/jokes/random");
      if (!response.ok) {
        return {
          content: [{ type: "text",
text: `No joke available at the moment.` }],
        };
      }
      const data = await response.json();
      return { content: [{ type: "text", text: data.value }] };
    } catch (err) {
      const message =
```

Initializes a new MCP server instance with a name and version

Defines a tool on the MCP server with a name, description, input schema, and handler function